

Section 1.6

Exercise (1.6.2). If $\varphi : G \rightarrow H$ is an isomorphism, prove that $|\varphi(x)| = |x|$ for all $x \in G$. Deduce that any two isomorphic groups have the same number of elements of order n for each $n \in \mathbb{Z}^+$. Is the result true if φ is only assumed to be a homomorphism.

Proof: Let $\varphi : G \rightarrow H$ be an isomorphism and suppose $x \in G$ such that $|x| = \alpha$. By properties of homomorphisms,

$$\varphi(x)^\alpha = \varphi(x^\alpha) = \varphi(e) = e.$$

Suppose there is a $\beta < \alpha$ such that $\varphi(x)^\beta = e$. By properties of homomorphisms it would follow that, $x^\beta = e$ which is clearly a contradiction.

Since φ is a bijection, by the previous result if there are m elements in G of order n there must be m elements in H of order n . This conclusion does not hold if φ is only a homomorphism, consider $\varphi : \mathbb{Z}_4 \rightarrow \mathbb{Z}_3$ such that $\varphi(x) = x \pmod 3$. There are two elements in $\mathbb{Z}_4$ of order 5, and none in $\mathbb{Z}_3$. $\square$

Exercise (1.6.3). If $\varphi : G \rightarrow H$ is an isomorphism, prove that G is abelian if and only if H is abelian. If $\varphi : G \rightarrow H$ is a homomorphism, what additional conditions on ϕ are sufficient to ensure that if G is abelian, then so is H .

Proof: Let $\varphi : G \rightarrow H$ be an isomorphism, and suppose that G is abelian. By definition for $a, b \in G$, $ab = ba$. Since φ is well defined, $\varphi(ab) = \varphi(ba)$ and by properties of homomorphisms, $\varphi(a)\varphi(b) = \varphi(b)\varphi(a)$. Since φ is a surjection every element in H can be written in terms of $\varphi(x)$ for some $x \in G$. Thus H is abelian.

Now suppose H is abelian. Since φ is a surjection, consider $\varphi(a), \varphi(b) \in H$, and note $\varphi(a)\varphi(b) = \varphi(b)\varphi(a)$. By properties of homomorphisms, $\varphi(ab) = \varphi(ba)$ and since φ is a bijection we can apply φ^{-1} to get, $ab = ba$. Thus G is abelian.

If $\varphi : G \rightarrow H$ is only a homomorphism, it is necessary for φ to also be a surjection. This is needed when we define or note, $\varphi(a), \varphi(b) \in H$ since every element in H must be able to be written in terms of $\varphi(x)$ for $x \in G$. This is awful yikers. $\square$

Exercise (1.6.5). Prove that the additive groups $\mathbb{R}$ and $\mathbb{Q}$ are not isomorphic.

Proof: Suppose for the sake of contradiction that $\phi : \mathbb{R} \rightarrow \mathbb{Q}$ is an isomorphism. By definition ϕ is also a bijection from $\mathbb{R} \rightarrow \mathbb{Q}$ and therefore $|\mathbb{R}| = |\mathbb{Q}|$. Thus a contradiction. $\square$

Exercise (1.6.6). Prove that the additive groups $\mathbb{Z}$ and $\mathbb{Q}$ are not isomorphic.

Proof: Suppose for the sake of contradiction that $\phi : \mathbb{Q} \rightarrow \mathbb{Z}$ is an isomorphism. By properties of isomorphisms we know that $\phi(1) = 1$. Now consider $a \in \mathbb{Z}$. by definition of $\mathbb{Z}$, $a = 1^a$. By substitution, and properties of homomorphisms $a = \phi(1)^a = \phi(1^a) = \phi(a)$. Therefore we have shown that $\phi(\mathbb{Z}) = \mathbb{Z}$ and thus ϕ cannot be an injection. $\square$

Exercise (1.6.14). Let G and H be groups and let $\varphi : G \rightarrow H$ be a homomorphism. Define the kernel of φ to be $\{g \in G | \varphi(g) = 1_H\}$. Prove that the kernel of φ is a subgroup of G .

Proof: Let G and H be groups and let $\varphi : G \rightarrow H$ be a homomorphism. Suppose $g \in \ker(\varphi)$.

By properties of isomorphisms, we know that $\varphi(1_G) = 1_H$. Consider that,

$$\varphi(1_G) = 1_H,$$

$$\varphi(g^{-1}g) = 1_H,$$

$$\varphi(g^{-1})\varphi(g) = 1_H,$$

$$\varphi(g^{-1}) = 1_H.$$

Thus $\ker(\varphi)$ is closed with respect to inverses.

Suppose $a, b \in \ker(\varphi)$. By properties of homomorphisms it follows that,

$$\varphi(a)\varphi(b) = 1_H,$$

$$\varphi(ab) = 1_H.$$

Therefore $\ker(\varphi)$ is closed. Thus $\ker(\varphi)$ is a subgroup of G . □

Prove that the kernel of φ is injective if and only if the kernel of φ is the identity subgroup of G .

Proof: Let G and H be groups and let $\varphi : G \rightarrow H$ be a homomorphism. Suppose that $\ker(\varphi)$ is injective. It follows that $|\ker(\varphi)| = 1$ since by definition $|\text{img}(\ker(\varphi))| = 1$. Note that,

$$\varphi(1_G) = \varphi(1_G 1_G),$$

$$\varphi(1_G) = \varphi(1_G)\varphi(1_G),$$

$$\varphi(1_G)^{-1}\varphi(1_G) = \varphi(1_G)^{-1}\varphi(1_G)\varphi(1_G),$$

$$1_H = \varphi(1_G).$$

Therefore $1_G \in \ker(\varphi)$. The other direction is trivial. □

Exercise (1.6.15). Define the map $\pi : \mathbb{R}^2 \rightarrow \mathbb{R}$ by $\pi((x, y)) = x$. Prove that π is a homomorphism and find the kernel of π .

Proof: Consider the map $\pi : \mathbb{R}^2 \rightarrow \mathbb{R}$ by $\pi((x, y)) = x$. Suppose $(a, b), (c, d) \in \mathbb{R}^2$. Note that

$$\pi((a, b) + (c, d)) = \pi((a + c, b + d)) = a + c = \pi((a, b)) + \pi((c, d)).$$

Note that $\ker(\pi) = \{(a, b) \in \mathbb{R}^2 | a = 0\}$. □

Exercise (1.6.17). Let G be any group. Prove that the map from G to itself defined by $g \rightarrow g^{-1}$ is a homomorphism if and only if G is abelian.

Proof: Let G be a group and $\varphi : G \rightarrow G$ such that $\varphi(g) = g^{-1}$. Suppose φ is a homomorphism. Consider $a, b \in G$, by properties of homomorphisms we know that,

$$ab = \varphi((ab)^{-1}) = \varphi(b^{-1}a^{-1}) = \varphi(b^{-1})\varphi(a^{-1}) = ba.$$

Thus G is abelian.

Now suppose that G is abelian. Consider $a, b \in G$,

$$\varphi((ab)) = (ab)^{-1} = b^{-1}a^{-1} = a^{-1}b^{-1} = \varphi(a)\varphi(b).$$

Thus φ is a homomorphism. □

Exercise (1.6.18). Let G be any group. Prove that the map from G to itself defined by $g \rightarrow g^2$ is a homomorphism if and only if G is abelian.

Proof: Let G be a group and $\varphi : G \rightarrow G$ such that $\varphi(g) = g^2$. Suppose φ is a homomorphism.

Consider $a, b \in G$, by properties of homomorphisms we know that,

$$(ab)(ab) = (ab)^2 = \varphi(ab) = \varphi(a)\varphi(b) = a^2b^2.$$

Applying a^{-1} on the left and b^{-1} on the right, we conclude that $ba = ab$. Thus G is abelian.

Now suppose that G is abelian. Consider $a, b \in G$,

$$\varphi((ab)) = (ab)^2 = abab = a^2b^2 = \varphi(a)\varphi(b).$$

Thus φ is a homomorphism. □

Exercise (1.6.19). Let $G = \{z \in \mathbb{C} \mid z^n = 1 \text{ for some } n \in \mathbb{Z}^+\}$. Prove that for any fixed integer $k > 1$ the map from G to itself defined by $z \rightarrow z^k$ is a surjective homomorphism but

is not an isomorphism.

Proof:

□

Section 1.7

Exercise (1.7.3). Show that the additive group $\mathbb{R}$ acts on the x, y plane $\mathbb{R}x\mathbb{R}$ by $r(x, y) = (x + ry, y)$

Proof: Let $g_1, g_2 \in \mathbb{R}$ and $(x, y) \in \mathbb{R}x\mathbb{R}$. Note that,

$$g_1(g_2(x, y)) = g_1(x + g_2y, y) = (x + g_2y + g_1y, y) = (x + (g_2 + g_1)y, y) = (g_1 + g_2)y.$$

Recall that in the additive group $\mathbb{R}$, 0 is the identity. Note,

$$0(x, y) = (x + 0y, y) = (x, y).$$

Thus the additive group $\mathbb{R}$ acts on $\mathbb{R}x\mathbb{R}$.

□

Exercise (1.7.4). Let G be a group acting on a set A and fix some $a \in A$. Show that the following sets are subgroups of G .

- a. The kernel of the action,

Proof: Let g, h be in the kernel of the action of G and $x \in A$. Consider gh acting on x , since g, h are in the kernel we get the following,

$$(gh) \cdot x = (g \cdot (h \cdot x)) = (g \cdot x) = x.$$

Thus gh is in the kernel of the action of G . Therefore the group is closed with respect to function composition. Note that,

$$g^{-1} \cdot (x) = g^{-1} \cdot (g \cdot (x)) = (g^{-1}g) \cdot (x) = 1 \cdot x = x.$$

Therefore g^{-1} is in the kernel of the action of G , therefore the group is closed with respect to inverses. Thus the kernel of the action of G is a subgroup of G . $\square$

b. $\text{Stab}_G(a) = \{g \in G \mid ga = a\}$.

Proof: Let $g, h \in \text{Stab}_G(a)$. Consider gh acting on a , since g, h are in the stabilizer group of a in G we get,

$$(gh) \cdot a = (g \cdot (h \cdot a)) = (g \cdot a) = a.$$

Thus $gh \in \text{Stab}_G(a)$. Therefore the group is closed with respect to function composition. Note that,

$$g^{-1} \cdot (a) = g^{-1} \cdot (g \cdot (a)) = (g^{-1}g) \cdot (a) = 1 \cdot a = a.$$

Therefore $g^{-1} \in \text{Stab}_G(a)$, therefore the group is closed with respect to inverses. Thus $\text{Stab}_G(a) \subseteq G$. $\square$

Exercise (1.7.6). Prove that a group G acts faithfully on a set A if and only if the kernel of the action is the set consisting only of the identity.

Proof: Suppose G , a group acting faithfully on the set A . Let g be in the kernel of the action of G . By definition for $a \in A$, we know that $g \cdot a = a$. Also note that $1 \cdot a = a$ and therefore g and 1 map to the same permutation representation. Since G is a faithful action, by definition $\varphi : G \rightarrow S_A$ is injective, and therefore $g = 1$. Thus kernel of the action of G is only the identity.

Suppose the kernel of the action is the set consisting only of the identity. For the sake of contradiction suppose that G does not act faithfully on A , and therefore there

exists $g, h \in G$ such that $g \neq h$ for all $a \in A$, $g \cdot a = h \cdot a$. Note that

$$g^{-1}h \cdot (a) = g^{-1} \cdot (h \cdot (a)) = g^{-1} \cdot (g \cdot (a)) = (g^{-1}g) \cdot (a) = 1 \cdot a = a.$$

Note $g^{-1}h \neq 1$ since $g \neq h$, therefore $g^{-1}h$ in the kernel of the action is a contradiction.

□

Exercise (1.7.13). Find the kernel of the left regular action.

Proof: Let G be a group, the kernel of its left regular action is the set of all $g \in G$ such that $ga = a$ for all $a \in G$. Clearly this set contains the identity element, and since G is a group we know that the identity element is unique. Thus the kernel of the left regular action is simply the identity. □

Exercise (1.7.14). Let G be a group and let $A = G$. Show that if G is non-abelian then the maps defined $g \cdot a = ag$ for all $g, a \in G$ do not satisfy the axioms of a left group action of G on itself.

Proof: Let G be non-abelian group and for the sake of contradiction suppose that the map $g \cdot a = ag$ for all $g, a \in G$ is a group action. By definition there exist $g, h \in G$ such that $gh \neq hg$. Note that

$$g \cdot (h \cdot a) = g \cdot (ah) = ahg$$

By properties of group actions,

$$g \cdot (h \cdot a) = gh \cdot (a) = agh.$$

Therefore we get, $ahg = agh$ and by left cancellation of a we get $hg = gh$ which is a contradiction. Thus $g \cdot a = ag$ does not satisfy the axioms of a left group action of G on itself. □

Exercise (1.7.15). Let G be any group and let $A = G$. Show that the maps defined by $g \cdot a = ag^{-1}$ for all $g, a \in G$ do satisfy the axioms of a left group action of G on itself.

Proof: Let G be any group and let $A = G$. Suppose the maps defined by $g \cdot a = ag^{-1}$ for all $g, a \in G$. Consider

$$(gh) \cdot a = a(gh)^{-1} = a(h)^{-1}(g)^{-1} = (g \cdot (h \cdot a)).$$

Also note,

$$1 \cdot a = a(1)^{-1} = a$$

Therefore $g \cdot a = ag^{-1}$ defines a group action. □

Exercise (1.7.16). Let G be any group and let $A = G$. Show that the maps defined by $g \cdot a = gag^{-1}$ for all $g, a \in G$ do satisfy the axioms of a left group action of G on itself.

Proof: Let G be any group and let $A = G$. Suppose the maps defined by $g \cdot a = gag^{-1}$ for all $g, a \in G$. Consider

$$(gh) \cdot a = (gh)a(gh)^{-1} = g(ha(h)^{-1})(g)^{-1} = (g \cdot (h \cdot a)).$$

Also note,

$$1 \cdot a = (1)a(1)^{-1} = a$$

Therefore $g \cdot a = gag^{-1}$ defines a group action. □

Exercise (1.7.17). Let G be a group and let G act on itself by left conjugation, so each $g \in G$ maps G to G by,

$$x \rightarrow gxg^{-1}$$

For fixed $g \in G$, prove that conjugation by g is an isomorphism from G into itself. Deduce that x and gxg^{-1} have the same order for all x in G and that for any subset A of G , $|A| = |gAg^{-1}|$.

Proof: Suppose $g \in G$, and let $\varphi : G \rightarrow G$ be defined by $\varphi(x) = gxg^{-1}$. Consider $a, b \in G$ and note that,

$$\varphi(a)\varphi(b) = g(a)g^{-1}g(b)g^{-1} = g(a)(g^{-1}g)(b)g^{-1} = g(ab)g^{-1} = \varphi(ab).$$

Therefore φ is a homomorphism.

Suppose $a, b \in G$ such that $\varphi(a) = \varphi(b)$. By left and right cancellation we get $a = b$ and therefore φ is an injection. Since φ maps from G to G it is also a surjection. Therefore φ is an isomorphism.

By properties of isomorphisms, namely what was proved in exercise 1.6.2 we know that for all $x \in G$, $|\varphi(x)| = |x|$ and therefore $|x| = |gxg^{-1}|$. It also follows that for any $A \subseteq G$, $\varphi : A \rightarrow gAg^{-1}$ is an isomorphism, and therefore $|A| = |gAg^{-1}|$. $\square$

Exercise (1.7.18). Let H be a group acting on a set A . Prove the relation $\sim$ on A defined by

$$a \sim b \text{ if and only if } a = h \cdot b \text{ for some } h \in H.$$

is an equivalence relation.

Proof: Consider $a \in A$. Clearly $a \sim a$ since, $a = 1 \cdot a$. Now suppose $b \in A$ such that $a \sim b$.

By definition there exists some $h \in H$ such that $a = h \cdot b$. Applying h^{-1} we get, $h^{-1} \cdot a = h^{-1} \cdot (h \cdot b) = (h^{-1}h) \cdot b = b$ and therefore $b \sim a$. Suppose $a, b, c \in A$ such that $a \sim b$ and $b \sim c$. By definition we get that there exists some $g, h \in H$ such that $a = h \cdot b$ and $b = g \cdot c$. By substitution we know that $a = g \cdot (h \cdot c) = (gh) \cdot c$ and therefore $a \sim c$. Thus $\sim$ is an equivalence relation. $\square$

Exercise (1.7.19). Let H be a subgroup of the finite group G and let H act on G by left multiplication. Let $x \in G$ and let O be the orbit of x under the action H . Prove the map

$$H \rightarrow O \text{ defined by } h \rightarrow h \cdot x$$

is a bijection. From this and the preceding exercise deduce Lagrange's Theorem.

Proof: Let $\varphi : H \rightarrow O$ define the map $h \rightarrow hx$. Suppose $g, h \in H$ such that $\varphi(g) = \varphi(h)$.

By definition we know that $gx = hx$ by right cancellation we conclude that $g = h$ and thus φ is an injection. Suppose $y \in O$. By definition there exists some $h \in H$

such that $y = hx$, and therefore $y = \varphi(h)$. Thus φ is a surjection.

Suppose a group G and subgroup H . By exercise 1.7.18 we know that the orbits under the action of H partition G . Suppose there are n orbits, so it follows that, $|G| = \sum_{i=1}^n |O_i|$. By the previous result since H is a subgroup we know that all orbits, have cardinality $|H|$. Thus it follows that $|G| = n|H|$ and finally $|H| \mid |G|$. $\square$

Exercise (1.7.20). Show that the group of rigid motions of a tetrahedron is isomorphic to a subgroup of S_4

Proof: Suppose G is the group of rigid motions of a tetrahedron. Let $A = [4]$ be the set of vertices. Now let G act on A and note that for each $g \in G$ we get $\varphi_g : [4] \rightarrow [4]$, defined by $g \cdot a = \varphi_g(a)$ with $\varphi_g \in S_4$. Note that each $g \in G$ induces a different permutation on a we know that the map from $G \rightarrow S_4$ defined $\sigma : g \rightarrow \varphi_g$ is an injection, and a homomorphism by properties of group actions. Finally consider $\sigma(G)$ a subset S_4 , which makes the map $G \rightarrow \sigma(G)$ defined $\sigma : g \rightarrow \varphi_g$ a surjection and therefore an isomorphism. $\square$

Section 2.1

Exercise (2.1.3a). Show that the $A = \{1, r^2, s, sr^2\}$ is a subgroup of the dihedral group $D_{2(4)}$.

Solution: Since $D_{2(4)}$ is finite it is sufficient to show that A is closed under composition for it to be a group. Recall the typical dihedral group presentation $D_{2(4)} = \{r, s \mid r^4 = s^2 = 1, rs = sr^{-1}\}$ and consider the following composition table,

	1	r^2	s	sr^2
1	1	r^2	s	sr^2
r^2	r^2	1	sr^2	s
s	s	sr^2	1	r^2
sr^2	sr^2	s	r^2	1

Clearly A is closed under composition, therefore it is a subgroup of $D_{2(4)}$. $\square$

Exercise (2.1.4). Give an explicit example of a group G and an infinite subset H of G that is closed under the group operation but is not a subgroup of G .

Solution: Any example must have the property that H is not closed under inverses. Consider $\mathbb{R}^*$ under multiplication, and the set $\mathbb{N}$. Clearly any product of two natural numbers is a natural number. The set also contains the multiplicative identity, and is an infinite subset of $\mathbb{R}$. However $\mathbb{N}$ is not closed under inverses, and is therefore not a subgroup of $\mathbb{R}^*$. $\square$

Exercise (2.1.5). Prove that G cannot have a subgroup H with $|H| = n - 1$, where $|G| = n > 2$.

Proof: Let G be a group with $|G| = n > 2$. Suppose H is a subgroup of G such that $|H| = n - 1$. Since G is finite and H is a subgroup by Lagrange's Theorem we know that $|H| \mid |G|$, and therefore for some $i \in \mathbb{Z}$, $|G| = |H|i$. By substitution we get, $n = (n - 1)i$ and therefore $i = n/(n - 1)$. Since $n > 2$, there is no integer i which satisfies $i = n/(n - 1)$, thus a contradiction. $\square$

Exercise (2.1.6). Let G be an abelian group. Prove that $A = \{g \in G \mid |g| < \infty\}$ is a subgroup of G .

Proof: Let G be an abelian group. Suppose $A = \{g \in G \mid |g| < \infty\}$. Clearly $1 \in A$. Let $a, b \in A$, by definition there exists some $m, n \in \mathbb{N}$ such that $|a| = m$ and $|b| = n$. Consider ab^{-1} , since G is abelian we know that,

$$(ab^{-1})^{mn} = (a^m)^n (b^n)^{m(-1)} = 1^n 1^m = 1.$$

Thus ab^{-1} must have finite order, and therefore $ab^{-1} \in A$. Thus A is a subgroup of G . $\square$

Exercise (2.1.8). Let H and K be subgroups of G . Prove that $H \cup K$ is a subgroup if and only if either $H \subseteq K$ or $K \subseteq H$.

Proof: Let H and K be subgroups of G . Suppose that $H \cup K$ is a subgroup and without loss of generality let $H \not\subseteq K$. Therefore there exists some $h \in H$ such that $h \notin K$. Consider $k \in K$ and note that $hk \in H \cup K$ and therefore $hk \notin K$ since $hk \in K$ implies $hk(k^{-1}) = h \in K$. Thus $hk \in H$. It then follows that $(h^{-1}h)k = k \in H$ and therefore $K \subset H$.

Without loss of generality suppose $K \subseteq H$. Therefore $H \cup K = H$, which is by definition a subgroup. $\square$

Section 2.2

Exercise (2.2.5). In each of the parts show that for the specified group G and subgroup A of G , $C_G(A) = A$ and $N_G(A) = G$.

a $G = S_3$ and $A = \{1, (123), (132)\}$.

Solution: $\square$

b $G = D_{2(4)}$ and $A = \{1, s, r^2, sr\}$.

Solution: $\square$

c $G = D_{2(5)}$ and $A = \{1, r, r^2, r^3, r^4\}$.

Solution: $\square$

Exercise (2.2.6). Let H be a subgroup G .

a Show that $H \leq N_G(H)$. Give an example to show that this is not necessarily true if H is not a subgroup.

Proof: Let H be a subgroup G . Suppose $h \in H$. Note that since H is a subgroup for all $a \in H$ we know that $hah^{-1} \in H$ and therefore by definition $h \in N_G(H)$ and thus $H \leq N_G(H)$.

Let $G = S_3$, $H = \{1, (123)(132)(12)\}$. Note that $(123)H(132) = \{1, (123)(132), (13)\} \neq H$ and therefore $(12) \notin N_G(H)$ so $H \not\leq G$. $\square$

b Show that $H \leq C_G(H)$ if and only if H is abelian.

Proof: Let H be a subgroup G . Suppose that $H \leq C_G(H)$. Consider $g, h \in H$ and note that since H is a subgroup of $C_G(H)$ we know that $ghg^{-1} = h$, therefore by right cancellation $gh = hg$. Thus H is abelian.

Now suppose H is abelian. Consider $g, h \in H$ and note that since $gh = hg$ we know that $ghg^{-1} = h$. Therefore $g \in C_G(H)$ and thus $H \leq C_G(H)$. $\square$

Exercise (2.2.7). Let $n \in \mathbb{Z}$ with $n \geq 3$. Prove the following:

a $Z(D_{2n}) = 1$ if n is odd.

Proof:

$\square$

b $Z(D_{2n}) = \{1, r^k\}$ if $n = 2k$.

Proof:

$\square$

Exercise (2.2.8). Let $G = S_n$, fix an $i \in [n]$ and let $G_i = \{\sigma \in G \mid \sigma(i) = i\}$. Use group actions to prove that G_i is a subgroup of G . Find $|G_i|$.

Proof: Let $G = S_n$, fix an $i \in [n]$ and suppose $G_i = \{\sigma \in G \mid \sigma(i) = i\}$. Clearly G_i is finite, therefore it suffices to show that G_i is nonempty and closed under function composition. Clearly $1 \in G_i$ by definition. Now consider $a, b \in G_i$, by the definition of our action G ,

$$(ab)(i) = a(b(i)) = a(i) = i.$$

Therefore $ab^{-1} \in G_i$ and G_i is closed under function composition thus G_i is a subgroup of G . $\square$

Exercise (2.2.9). For any subgroup H of G and any nonempty subset A of G define $N_H(A)$ to be the set $\{h \in H \mid hAh^{-1} = A\}$. Show that $N_H(A) = N_G(A) \cap H$ and deduce that $N_H(A)$ is a subgroup of H .

Proof: Let H be a subgroup of G and for any nonempty subset A of G suppose $N_H(A)$ to be the set $\{h \in H \mid hAh^{-1} = A\}$. Since H is a subgroup of G we know that $N_H(A) \leq N_G(A)$ and $N_H(A) \leq H$ then it follows that $N_H(A) = N_G(A) \cap H$. Let $g \in N_G(A) \cap H$, by definition $gAg^{-1} = A$ and $g \in H$. Thus $g \in N_H(A)$. Therefore

$$N_H(A) = N_G(A) \cap H.$$

Clearly $N_H(A)$ is nonempty since $1A1 = A$. Also $N_H(A)$ is closed with respect to inverses, for $h \in N_H(A)$ we know that $hAh^{-1} = A$. By left and right cancellation we get, $A = h^{-1}Ah$ therefore $h^{-1} \in N_H(A)$. Finally $N_H(A)$ is closed with respect to composition, let $a, b \in N_H(A)$ and consider that $abA(ab)^{-1} = a(bAb^{-1})a^{-1} = a(A)a^{-1} = A$ therefore $ab \in N_H(A)$. Thus $N_H(A)$ is a subgroup. $\square$

Exercise (2.2.10). Let H be a subgroup of order 2 in G . Show that $N_G(H) = C_G(H)$. Deduce that if $N_G(H) = G$ then $H \leq Z(G)$.

Proof:

$\square$

Exercise (2.2.11). Prove that $Z(G) \leq N_G(A)$ for any subset A of G .

Proof: Let A be a subset of G . Suppose $g \in Z(G)$. By definition $gx = xg$ for all $x \in G$.

By right cancellation it follows that $gxg^{-1} = x$ for all $x \in G$. Since A is a subset of G , it follows that $gag^{-1} = a$ for all $a \in A$ and therefore $g \in N_G(A)$. Thus $Z(G) \leq N_G(A)$. $\square$